



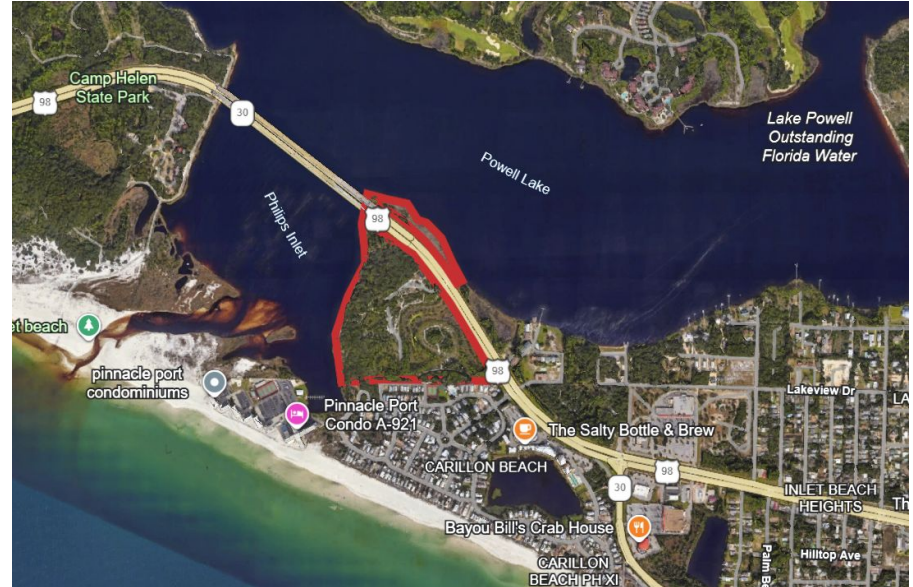
# Wellness Resort and Eco Spa Sustainability Master Plan

Sunnyside, FL



# Site Background

- Florida Panhandle
  - 10 minutes from Panama City Beach
- ~41 acres
- Elevation Estimate
  - Min: 2 ft | Median: 6 ft | Max: 20 ft
- Next to Powell Lake



# Site Master Plan

## Key

- R Resort  
Amenities (first floor) -- approx. 20,000 sqf  
60 Keys
- I Medical Institute  
approx. 106,000 sqf
- M Lakefront Manoir  
8 units
- LM Lakefront Maison  
2 buildings with 3 units each (total 6 units)
- GM Garden Maison  
4 buildings with 4-6 units each (total 20 units)
- PT1 Pied-à-terre 1  
Amenities (first floor) -- approx. 31,000 sqf  
16 or 18 units (2nd and 3rd floors)  
6 Penthouse units
- PT2 Pied-à-terre 2  
Amenities (Institute) - approx. 6,500 sqf  
21-24 units (2nd and 3rd floor)  
6 ~ 7 Penthouse units  
Back of house approx. 13,000 sqf
- PT3 Pied-à-terre 3  
Amenities -- approx. 12,500 sqf  
41 to 43 units (2nd and 3rd floor)  
6 to 7 Penthouse units
- PT4 Pied-à-terre 4  
15 ~ 20 units  
3 ~ 4 Penthouse units
- PM Peninsula Maisonette  
9 ~ 11 units  
TBC upon receipt of new wetland survey
- A Back of House
- B Gatehouse
- C Boardwalk
- D Gardens
- Existing Trees
- Proposed Trees
- Wetland Buffer 2001
- Wetland Buffer 2016
- Indicative Extent of Basement (to be confirmed)



# Master Sustainability Plan

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Context

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Resiliency

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Energy Plan

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Transportation Plan

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Water Plan

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Summary



# Resiliency Master Plan



# Resiliency Issues

- Hurricanes
- Flooding and Sea Level Rise
- Wildfires and Air Quality



<https://www.newsherald.com/picture-gallery/news/2020/09/15/hurricane-sally-floods-roads-and-brings-rain-panama-city-beach/5806068002/>

# Hurricanes and Mitigation Strategies

- Continuous Load Path
- Impact-Rated Envelopes
- Hardened Roofing



# Flooding and Sea Level Rise



Expected sea level rise in 2060 (left) and in 2100 (right)

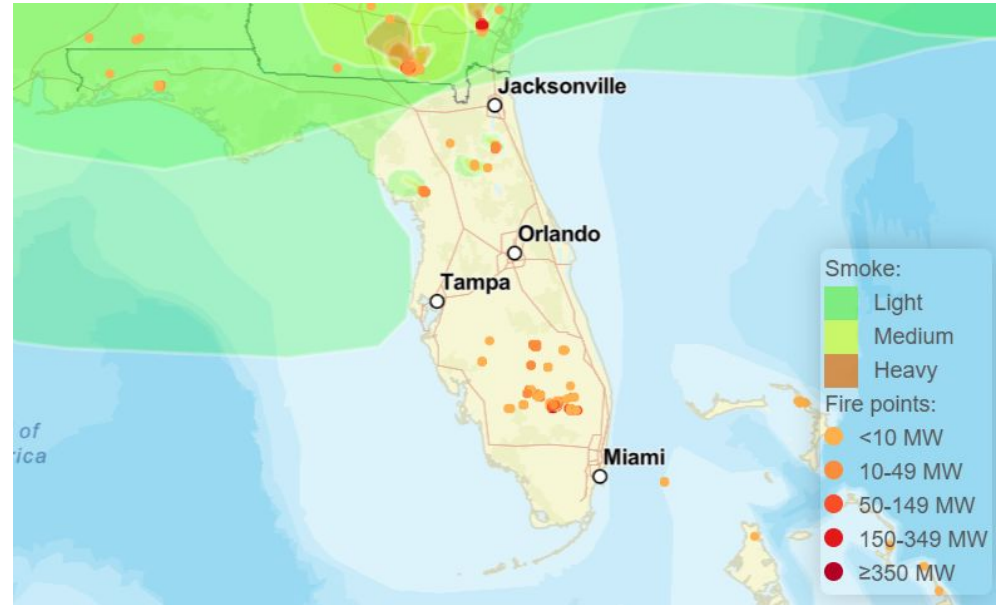
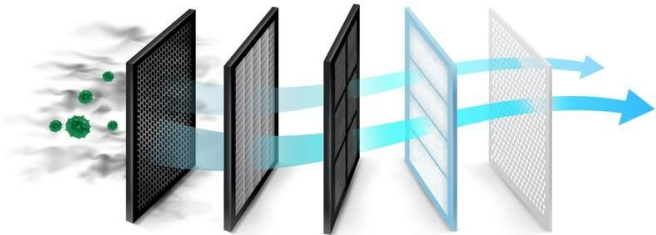
# Flooding and Sea Level Rise Mitigations

- Elevated Infrastructure
- Utility Protection
- Shoreline Reinforcement
- Deployable Barriers



# Wildfires and Air Quality Mitigations

- Defensible Space
- Advanced Filtration
- Real-time Monitoring





# Energy Master Plan



# Energy Demand

| Category             | Building Type           | Quantity | Number of Floors | Unit Area (sq. ft.) | Total Area (sq. ft.) | Roof Area (sq. ft.) |
|----------------------|-------------------------|----------|------------------|---------------------|----------------------|---------------------|
| Resort               | Lodging - Large Hotel   | 1        | 1                | 20,000              | 20,000               | 20,000              |
| Medical Institute    | Healthcare - Outpatient | 1        | 2                | 106,000             | 106,000              | 53,000              |
| Lakefront Manoir     | Apartment - Mid Rise    | 8        | 2                | 2,500               | 20,000               | 10,000              |
| Lakefront Maison     | Apartment - Mid Rise    | 2        | 2                | 2,500               | 5,000                | 2,500               |
| Garden Maison        | Apartment - Mid Rise    | 2        | 1                | 12,600              | 25,200               | 12,600              |
| Pied-a-terre 1       | Apartment - Mid Rise    | 1        | 4                | 31,000              | 124,000              | 31,000              |
| Pied-a-terre 2       | Apartment - Mid Rise    | 1        | 4                | 24,690              | 24,690               | 6,173               |
| Pied-a-terre 3       | Apartment - Mid Rise    | 1        | 4                | 25,318              | 25,318               | 6,330               |
| Pied-a-terre 4       | Apartment - Mid Rise    | 1        | 3                | 10,000              | 10,000               | 3,333               |
| Peninsula Maisonette | Apartment - Mid Rise    | 11       | 1                | 1,500               | 16,500               | 16,500              |
| Parking Structures   | Parking                 | 1        | 2                | 68,400              | 136,800              | 68,400              |

# Energy Demand

Used ASHRAE Building Energy Code data

| Category                | Source EUI (kBtu/ft2-year) | Total Area (sq. ft.) | Energy usage (kBtu/year) | Energy Usage (kWh/year) |
|-------------------------|----------------------------|----------------------|--------------------------|-------------------------|
| Lodging - Large Hotel   | 164.8                      | 20,000               | 3,296,000                | 965,962                 |
| Healthcare - Outpatient | 228.9                      | 106,000              | 24,263,400               | 7,110,899               |
| Apartment - Mid Rise    | 103.6                      | 250,708              | 25,973,349               | 7,612,035               |
| <b>Totals</b>           |                            | <b>376,708</b>       | <b>53,532,749</b>        | <b>15,688,896.22</b>    |

# PV System

NPV = \$2,193,177.65 net savings

Payback period: 10 years

~20% of annual building load

| kWh/year/kw | kW/sqft | kWh/year/sqft | System size (kW) | Total roof area (sqft) | Area roof used for pv (sqft) | Yearly PV generation (kWh) | Yearly building load (kW) |
|-------------|---------|---------------|------------------|------------------------|------------------------------|----------------------------|---------------------------|
| 1,505.00    | 0.015   | 22.58         | 2421.53          | 161,435.33             | 161,435.33                   | 3,644,402.65               | 15,688,896.22             |

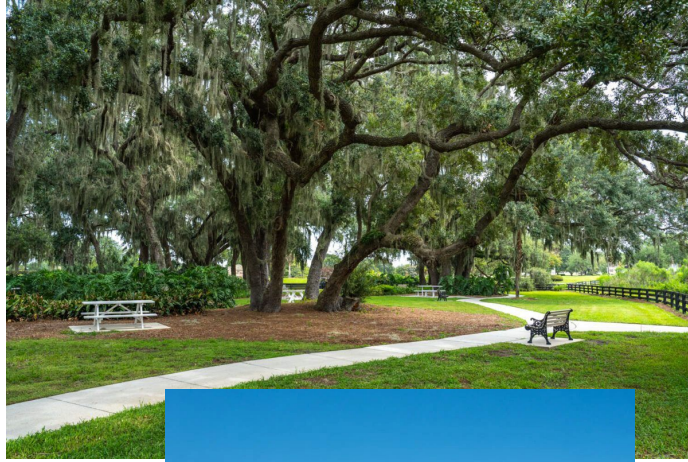


# Transportation Master Plan



# Options for Guests

- Walking
- Electric Shuttle
- Electric Golf carts



# Costs

|                | Installation Cost (\$) | Energy Cost (\$/yr) | Carbon Cost (\$/yr) | O&M (\$/yr) | Total Cost (\$/yr) | NPV (30yr)   |
|----------------|------------------------|---------------------|---------------------|-------------|--------------------|--------------|
| Base case      | 1,398,000.00           | 16,502.13           | 51,190.29           | 13,980.00   | 81,672.42          | 2,653,505.25 |
| Walking        | 2,398,000.00           | 0.00                | -1,596.00           | 100,000.00  | 98,404.00          | 3,910,710.67 |
| Shuttle only   | 1,858,000.00           | 10,117.80           | 0.00                | 300,000.00  | 310,117.80         | 6,625,270.69 |
| Golf cart only | 3,498,000.00           | 2,213.27            | 0.00                | 105,000.00  | 107,213.27         | 5,146,130.72 |
| Combination    | 3,128,000.00           | 6,112.84            | -1,596.00           | 202,500.00  | 207,016.84         | 6,310,356.20 |

# Recommendation

- Combination allows for:
  - Freedom for guests
  - Easy access to amenities
  - Comfort and wellbeing
  - Reduced emissions





# Water Master Plan



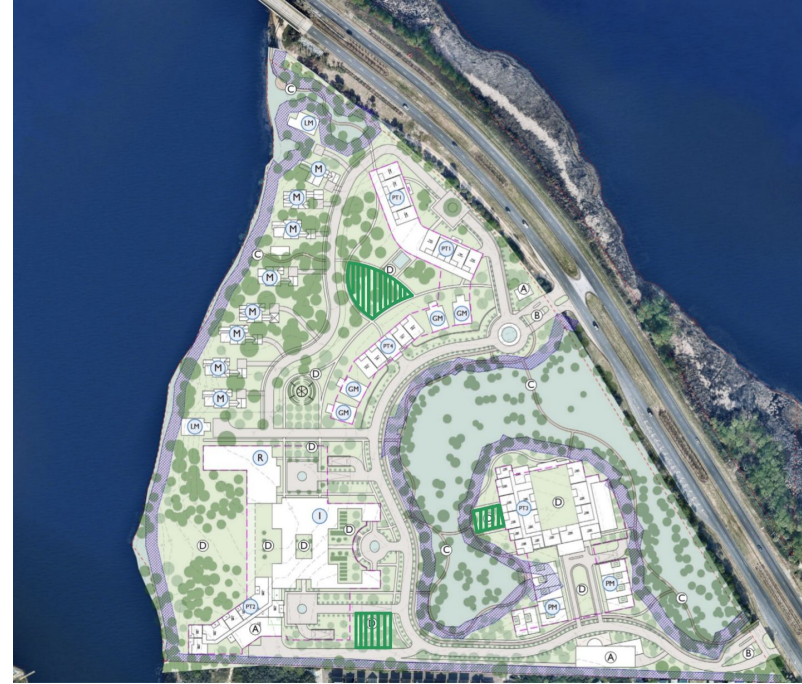
# Net Zero Water

- Reduce water usage
  - High efficiency fixtures
  - Rainwater capture from all roofs
- Reuse water
  - Capture gray water from showerheads and faucets
- Environmental responsibility
  - Constructed wetland
  - Reduce runoff



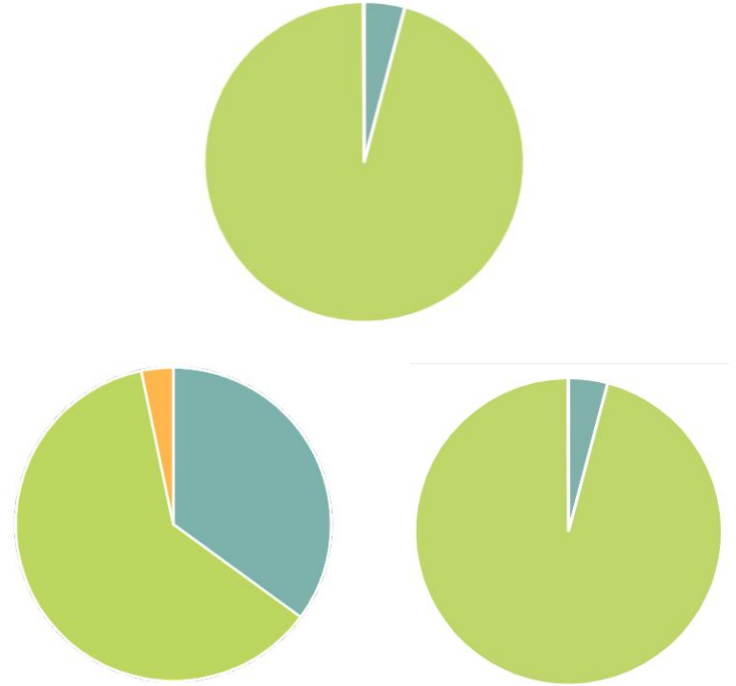
# Constructed Wetlands

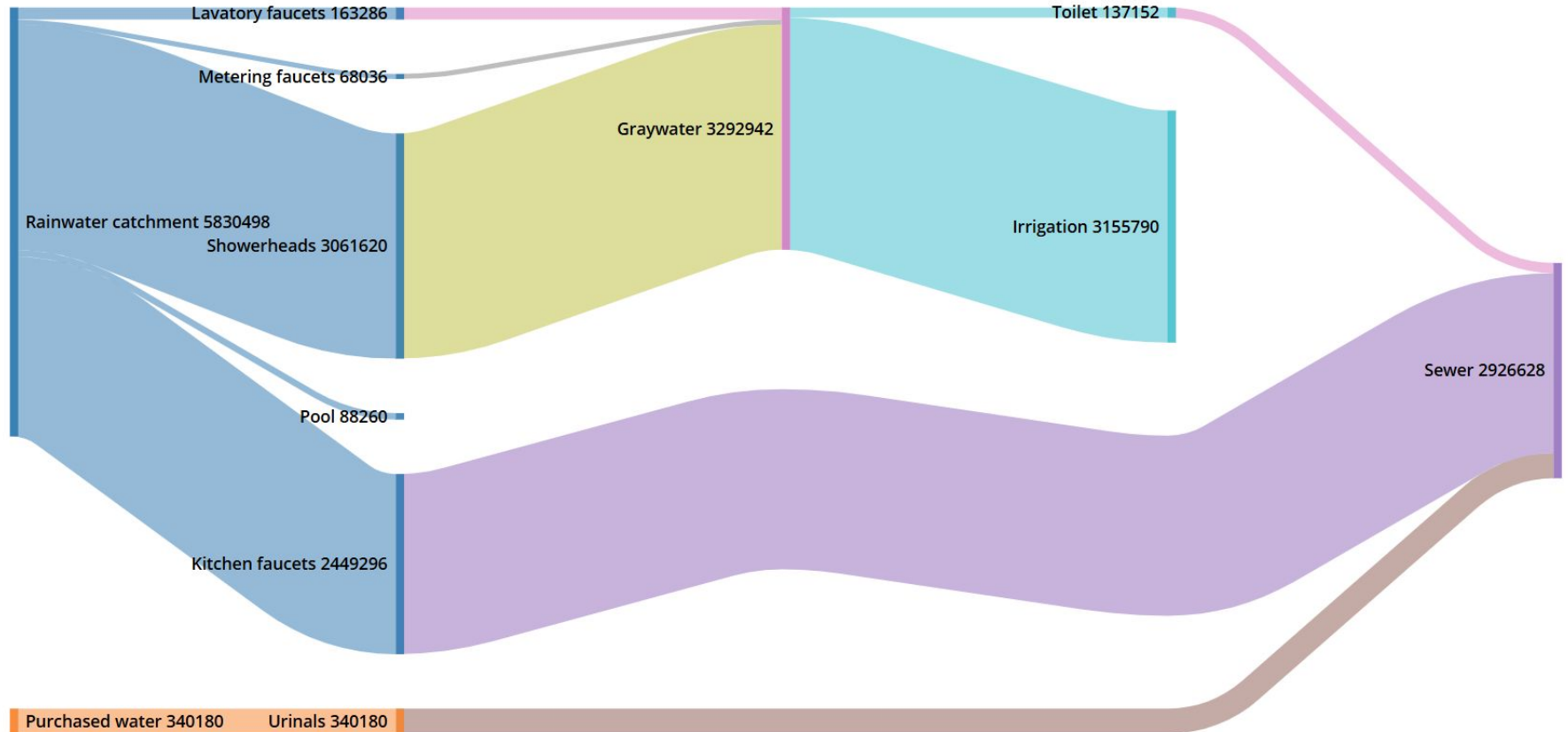
- Natural wetlands on-site are protected
  - 4200 total sq. ft of wetland spread across 3 wetlands of 1600, 1000 and 1800 sq. ft
- Used dark graywater
- Settling tank
- Particulate filter
- Light chlorine treatment



# Stormwater Calculations

- Sensitive coastal dune lake ecosystem nearby
- Three scenarios: pre development, base case and LID controls (rainwater harvesting, infiltration basins, permeable pavement)





# Summary

- The Wellness Resort and Eco Spa is a 41-acre site in Sunnyside, FL.
- The resiliency plan addresses hurricanes, flooding, and sea level rise.
- Mitigations include elevated infrastructure, hardened roofing, and defensible space.
- The PV system offers net savings with a 10-year payback period.
- Water conservation is achieved through high-efficiency fixtures and constructed wetlands.